

Spatial Clustering of Cultural and Natural Heritage Sites in Bosnia and Herzegovina Using DBSCAN

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GitHub: <https://github.com/dzankk/geo-tourism-clustering-dbscan>

Abstract—This seminar paper applies the DBSCAN algorithm to geographic data to analyze regional tourism in Bosnia and Herzegovina. The study uses a unique, self-collected dataset consisting of 272 historical, cultural, and natural heritage sites across the country. Because geographic coordinates sit on a sphere, the Haversine metric is used to compute accurate real-world distances between sites. While traditional algorithms like K-means force data into rigid circles, DBSCAN is used here to discover natural, arbitrarily shaped tourism corridors. Furthermore, the algorithm successfully separates dense tourism hubs from isolated noise points. Instead of discarding this noise as errors, this paper analyzes these outliers as “hidden gems”—isolated locations with untapped potential for sustainable rural eco-tourism. Ultimately, this project demonstrates how density-based machine learning can help regional planners prevent over-tourism in crowded cities and distribute economic benefits to underdeveloped rural areas.

Sažetak (Bosanski)—Ovaj seminarski rad primjenjuje DBSCAN algoritam na geografske podatke u svrhu analize regionalnog turizma u Bosni i Hercegovini. Rad koristi jedinstven, vlastiti skup podataka koji se sastoji od 272 kulturno-historijska i prirodna lokaliteta širom zemlje. Budući da se geografske koordinate nalaze na sferi, koristi se Haversine distanca za tačan proračun stvarnih udaljenosti između lokacija. Dok tradicionalni algoritmi poput K-means-a forsiraju krute kružne oblike, DBSCAN se ovdje koristi za otkrivanje prirodnih turističkih koridora proizvoljnog oblika. Pored toga, algoritam uspješno odvaja guste turističke centre od izolovanih tačaka koje detektuje kao šum. Umjesto da se taj šum odbaci kao greška, u ovom radu se te izolovane tačke analiziraju kao “skriveni dragulji” s neiskorištenim potencijalom za održivi ruralni eko-turizam. Na koncu, ovaj projekat pokazuje kako metode mašinskog učenja zasnovane na gustoći mogu pomoći u rješavanju problema prekomjernog turizma u preopterećenim gradovima i usmjeravanju ekonomskih benefita ka manje razvijenim ruralnim područjima.

Index Terms—DBSCAN, Spatial Clustering, Haversine Metric, Sustainable Tourism, Regional Planning, Bosnia and Herzegovina, BiH

I. INTRODUCTION

The strategic distribution and analysis of tourism destinations in Bosnia and Herzegovina (BiH, Bosnian: Bosna i Hercegovina) is highly important for regional development. While main urban centers and popular destinations often

face infrastructure pressure from seasonal over-tourism, many valuable historical, cultural, and natural heritage sites in rural areas remain isolated and disconnected from major travel networks. Traditional geometric methods for spatial classification usually rely on rigid administrative boundaries (such as entity or municipality borders) or use standard centroid-based partitioning metrics like K -means clustering.

However, real-world tourism distribution and travel patterns do not follow static linear borders or perfectly circular shapes. Instead, cultural and natural assets are naturally distributed along geographical features like river valleys, mountain passes, and historic roads, forming irregular spatial patterns of arbitrary shapes. While partitioning algorithms like k -means or k -medoids are common for data grouping, they have two main disadvantages in geographic data mining. First, their logic splits space into convex shapes resembling Voronoi diagrams, which cannot capture elongated or irregular clusters [1]. Second, they require the user to specify the target number of clusters k in advance, which is difficult when exploring an unknown spatial landscape [1].

To solve these limitations, density-based algorithms like DBSCAN (English: Density-Based Spatial Clustering of Applications with Noise) have become a standard choice in unsupervised machine learning and spatial data analysis [2]. DBSCAN identifies clusters by checking the local density of points, making it highly effective at finding clusters of arbitrary shapes and separating them from sparse background noise. Over the years, several advanced variations have been developed to improve on its basic framework. For example, OPTICS (English: Ordering Points To Identify the Clustering Structure) and LSDBC (English: Locally Scalable Density-Based Clustering) were introduced to handle hierarchical cluster distributions with varying densities, while DENCLUE (English: DENsity-based CLUsterinG) uses kernel density functions for smoother estimations [2]. Additionally, frameworks like HDBSCAN* (English: Hierarchical Density-Based Spatial Clustering of Applications with Noise) optimize the process by focusing strictly on core points to maintain better structural consistency across different density levels [2].

In geographic research and tourism mapping, analyzing how assets are distributed across a country is crucial for sustainable planning. Traditional approaches often rely on basic municipal

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registries or paper surveys, which are limited in capturing broader spatial relations [3]. Modern data science allows the collection of decentralized geospatial metadata, providing clean coordinates for tourism trajectory mining and cluster analysis over wide geographic landscapes [3].

However, applying standard spatial algorithms to unvetted geospatial data can lead to issues if the coordinates are highly concentrated in small areas due to data collection bias [4]. If a clustering engine treats every raw point with equal weight without validation, it might generate false-positive clusters that do not represent true public interest or real attraction corridors [4].

To address these spatial analysis challenges, recent studies frequently compare distance hierarchies and density variations within GIS (English: Geographic Information System) infrastructures to find performance gaps [3]. Furthermore, integrating unvetted spatial inventories without proper filtering can result in fragmented and unreliable regional models [5]. By focusing on objective density assessment and clear parameter calibration, this paper utilizes GIS and NLP (English: Natural Language Processing) research data frameworks to establish a reliable and clean computational workflow to map and analyze the spatial layout of cultural and natural heritage sites across Bosnia and Herzegovina.

II. USED ALGORITHM

The computational engine implemented throughout this research is the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. This machine learning methodology relies on dense spatial point neighborhoods to distinguish clustered groups from sparse background outliers without requiring prior knowledge regarding the total number of expected target segments.

A. Algorithmic Principles and Search Mechanics

The execution of DBSCAN requires two primary hyperparameters: the maximum neighborhood radius (ϵ) and the minimum number of points required to form a dense cluster (*MinPts*). Given a spatial dataset D , the algorithm scans the data points and categorizes each into one of three structural classes based on local density. The fundamental logic of DBSCAN relies on these density metrics to determine whether a point belongs to a cluster or should be classified as noise [1]. Rather than treating all points within a cluster equally, the algorithm distinguishes between internal elements (“core points”) and peripheral elements (“border points”), since the ϵ -neighborhood (N_ϵ) of a border point contains fewer points than that of a core point [1].

- **Core Points:** A coordinate point $p \in D$ is formally classified as a core point if its immediate ϵ -neighborhood contains a total number of data points equal to or greater than the user-specified *MinPts* threshold. This neighborhood zone is defined mathematically as:

$$N_\epsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \epsilon\} \quad (1)$$

where $\text{dist}(p, q)$ represents the geodesic distance function.

- **Border Points:** A coordinate point $r \in D$ is designated as a border point if it fails to fulfill the standalone density requirements of a core point, but falls within the immediate $N_\epsilon(p)$ neighborhood radius of an active, verified core point p .
- **Noise Points:** Any independent coordinate point that satisfies neither the core nor the border point density criteria is automatically classified as spatial noise ($c = -1$).

By relying strictly on localized geometric connectivity rules, the propagation mechanism of DBSCAN effectively identifies contiguous chains of density. A point q is defined as directly density-reachable from a point p if $q \in N_\epsilon(p)$ and p is a core point. Extending this concept, a point q is density-reachable from a point p if there exists a structural chain of intermediate points $p_1, p_2, \dots, p_n$ (where $p_1 = p$ and $p_n = q$) such that each p_{i+1} is directly density-reachable from p_i . Finally, two points p and q are considered density-connected if there exists an operating core point o such that both p and q are density-reachable from o . This relational mapping allows the algorithm to grow organic, winding cluster systems that follow topography, rivers, or valley roads rather than forcing circular boundaries.

B. Geospatial Coordinate Transformation via the Haversine Metric

Standard implementations of distance algorithms natively utilize the traditional Euclidean distance metric. However, executing Euclidean distance calculations on coordinate pairs (latitude, longitude) introduces structural spatial distortion because it assumes a perfectly flat two-dimensional plane, entirely neglecting the ellipsoidal curvature of the Earth. To solve this geometric limitation, the input coordinate matrices are projected into spherical space. Spatial distance vectors are computed directly using the central arc angle separating coordinate pairs on a sphere via the Haversine formula.

Prior to running the model, all spatial degrees are converted directly into spherical radians:

$$\phi = \text{latitude} \times \frac{\pi}{180}, \quad \lambda = \text{longitude} \times \frac{\pi}{180} \quad (2)$$

The real-world surface distance d separating two distinct geographic points on the sphere is calculated using the following equation:

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right) \quad (3)$$

where R represents the designated mean radius of the Earth, standardized within the execution engine to exactly 6371.009 km. To maintain strict mathematical alignment across the model, the user-defined parameter ϵ (expressed in kilometers) is divided by the Earth’s radius R prior to execution, generating a native radian distance threshold used directly by the underlying spatial engine.

The widely propagated claim that DBSCAN operates with an average-case time complexity of $\mathcal{O}(n \log n)$ has been heavily scrutinized and re-evaluated in modern database literature.

This logarithmic bound is not a native algorithmic property; rather, it hinges entirely on the architectural assumption that spatial indexing mechanisms—such as R^* -trees, k -d trees, or cover trees—can execute range search queries (RangeQuery) within an average time of $\mathcal{O}(\log n)$ [2]. By querying spatial boundaries using multidimensional access structures such as R^* -trees, the computational overhead of evaluating individual point neighborhoods scales to an average run-time complexity of $\mathcal{O}(n \log n)$ for a database containing n spatial coordinates [1]. When a dataset cannot be accelerated by an index structure, the performance scales quadratically to $\Theta(n^2 \cdot D)$, where D represents the computational cost of the distance metric [2].

Reductions from Hopcroft-hard geometric challenges—specifically the Unit-Spherical Emptiness Checking (USEC) and Bichromatic Closest Pairs (BCP) problems—prove that no universal spatial index can guarantee a strict $\mathcal{O}(\log n)$ query time across arbitrary data distributions without a computationally prohibitive preprocessing step [2]. Consequently, for spatial databases operating in a dimensionality of $d \geq 3$, structural configurations exist where index traversal slows down significantly, enforcing a foundational theoretical lower bound of $\Omega(n^{4/3})$ for the Euclidean density problem [2].

C. Automated Border Proximity Analysis

Geospatial research bound to a single country is naturally susceptible to edge-truncation bias. A dense cluster of historical landmarks located in close proximity to an international border may be incorrectly isolated or misclassified as spatial noise simply because cross-border data points are absent from the local input file. To address this data constraint, an automated boundary proximity function measures the distance between each coordinate and the approximate bounding limits of the country:

$$d_{\text{border}} = \min(|45.27 - \text{lat}|, |\text{lat} - 42.56|, |\text{lon} - 15.72|, |19.62 - \text{lon}|) \quad (4)$$

Coordinates matching a boundary threshold criteria of $\leq 0.25^\circ$ (approximately 27 km) are automatically flagged with an edge-truncation warning alert inside the map visualization layers, informing users that the observed density pattern may be artificially limited by geopolitical data boundaries.

III. SIMULATION RESULTS

A. Software Pipeline and Architecture Guide

The companion software solution is constructed modularly using the Python programming language [6]. Rather than combining data engineering, mathematical validation, and web deployment into a single file, the pipeline is split across separate functional modules:

- **data/heritage_sites_bih.csv**: Contains 272 unique historical records compiled from regional registries, storing structural variables including coordinates, municipality labels, categories, descriptions, and historical fun facts [7]. This original dataset was unified, parsed, and curated from primary municipal archives before being

uploaded publicly to Kaggle to support reproducible geospatial benchmarking.

- **src/data_processing.py**: Handles file loading, coordinate radian transformation, and border proximity calculation.
- **src/clustering_engine.py**: Manages the execution parameters of the DBSCAN model across multiple simultaneous scales.
- **notebooks/notebook.ipynb**: Handles optimization loops, silhouette metrics, and exports standalone HTML visual maps.
- **app.py**: Houses the interactive Streamlit user dashboard, enabling live parameter scaling using frontend sliders [8].

Structural anomalies during spatial exploration can manifest as a direct byproduct of low numerical resolution or discrete coordinate representations within the source dataset. When spatial attributes possess low precision, uniform coordinate gaps generate identical distance vectors; as a consequence, incremental expansions of the ϵ threshold can trigger abrupt, artificial cascading connections that merge independent local components into a single, uninformative mega-cluster [2]. To illustrate the underlying processing logic used to secure data integrity, the implementation block utilized to clean data and calculate boundary markers is outlined below:

```
def get_distance_to_border(lat, lon):
    dist_north = abs(45.27 - lat)
    dist_south = abs(lat - 42.56)
    dist_west = abs(15.72 - lon)
    dist_east = abs(19.62 - lon)
    return min(dist_north, dist_south, dist_west, dist_east)

df['border_distance_deg'] = df.apply(lambda r:
    get_distance_to_border(r['latitude'], r['longitude']), axis=1)
df['is_border_truncated'] = df['border_distance_deg'] <= 0.25
```

Listing 1. Data processing and border proximity engine snippet

B. Hyperparameter Optimization via Unsupervised Validation

Because true historical or economic labels do not exist beforehand for regional tourism networks, cluster validation cannot rely on traditional accuracy scores. Instead, parameter sensitivity is measured using the unsupervised Silhouette Coefficient (s). For each individual landmark i , the score is computed as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (5)$$

where $a(i)$ represents the mean intra-cluster distance between asset i and all other points belonging to the same cluster, and $b(i)$ represents the mean distance from asset i to all points located within the nearest neighboring cluster.

To streamline input overhead when identifying regional clusters in two-dimensional point sets, the foundational density parameters can be structurally narrowed down. Empirical evaluations show that sorted k -distance graphs do not diverge

significantly for values of $k > 4$; hence, the *MinPts* parameter can be reliably set to a constant value of 4 for 2D geographic coordination tasks, isolating ϵ as the primary volatile hyperparameter to calibrate [1].

In density-based spatial configurations, hyperparameter calibration exhibits asymmetric volatility. While variations in the *MinPts* parameter produce highly stable results with marginal structural impact on the overall cluster distribution, the selection of the neighborhood radius ϵ demands strict containment [2]. Empirical diagnostics indicate that lower numeric ranges for ϵ consistently capture authentic spatial structures, whereas inflated boundaries risk over-mapping data populations [2]. A parameter grid scan was executed across the input data matrix to discover the optimal value for ϵ , maintaining *MinPts* = 4 across testing iterations. The performance metrics are recorded in Table I.

TABLE I
DBSCAN SENSITIVITY ANALYSIS AND PERFORMANCE MATRIX

Epsilon (ϵ) in KM	Valid Clusters Formed	Mean Silhouette Score
10.0	18	0.2230
15.0	16	0.3249 (Mathematical Peak)
20.0	10	0.0126
25.0	7	0.0374
30.0	3	0.0177
40.0	1	N/A (Inherent Density Limit)
50.0	1	N/A (Inherent Density Limit)

To ensure that spatial partitioning provides meaningful structural insights, the resulting configuration must be evaluated against two fundamental degenerate criteria or “red flags.” First, an optimized DBSCAN deployment should isolate a clear noise percentage, typically bounded between 1% and 30% of the aggregate dataset [2]. Second, the relative volume of the largest discovered cluster must be strictly monitored; if the single largest component encompasses more than 20% to 50% of the total clustered points, it indicates that the ϵ parameter is overly permissive [2].

The numerical metrics in Table I indicate that peak spatial silhouette density occurs at exactly $\epsilon = 15.0$ km, separating the landmarks into 16 highly dense local destination pockets. Beyond $\epsilon = 40.0$ km, the silhouette evaluation becomes invalid because the single largest component swallows 100% of the data points, triggering a clear algorithmic red flag. Using these optimized parameters, neighborhood queries are executed efficiently within our pipeline. This ensures that the interactive Streamlit dashboard can recompute and render the maps quickly when the user changes the hyperparameters. Utilizing optimized tree-based indexing structures yields a 10-to-1000-fold runtime acceleration compared to a brute-force linear scan, while simultaneously outperforming modern approximate grid-based clustering frameworks [2].

C. Multi-Scale Analysis and Visualization Interpretation

The interactive software application, built using the Streamlit framework, maps and exports three separate density scales to a unified interface. This deployment allows regional planners to analyze spatial layouts dynamically using frontend sliders, dropdown menus for specific site isolation, and interactive map layers.

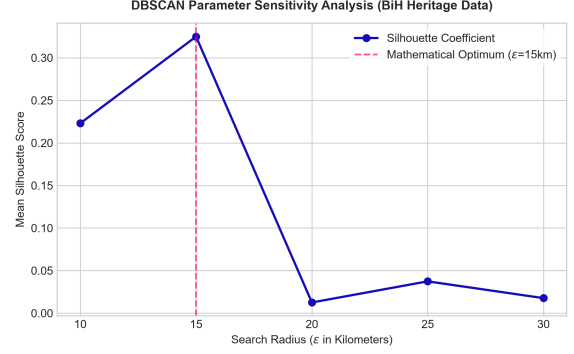


Fig. 1. Mean Silhouette Coefficient plotted across testing radii, showing the optimization path with a distinct mathematical peak at 15 km.

The relationship between search neighborhood dimensions and cluster boundaries is plotted in Figure 1. The clear mathematical peak at $\epsilon = 15.0$ km represents the optimal spatial partition for localized assets. At this level, the algorithm successfully matches the true localized clusters of heritage sites without blending independent historical regions together. When ϵ is too small (e.g., 10.0 km), clusters fragment heavily into tiny isolated nodes, reducing the global silhouette performance. Conversely, when ϵ expands past the 20.0 km threshold, distinct clusters merge prematurely, dropping the silhouette coefficient close to zero.

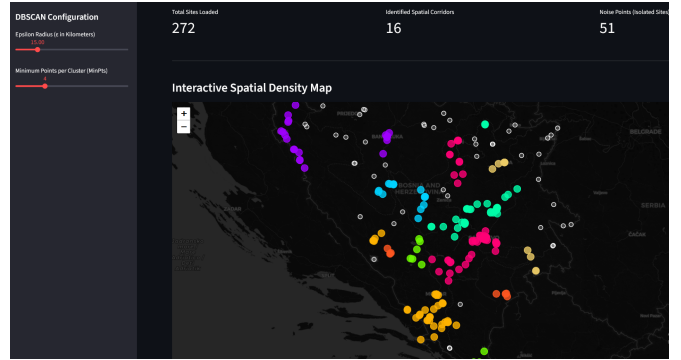


Fig. 2. Interactive dashboard layout displaying the optimized local framework ($\epsilon = 15.0$ km, *MinPts* = 4) with all 16 localized asset clusters exposed.

Figure 2 illustrates the baseline map interface demonstrating this optimized local framework ($\epsilon = 15.0$ km, *MinPts* = 4), exposing all 16 localized asset clusters across the country.

To inspect specific locations without obscuring adjacent data distributions, the interface includes a localized search dropdown menu and dynamic popup cards (Figure 3). When a user



Fig. 3. Detailed view of the user interface isolating a localized cluster via the sidebar dropdown, with an active interactive popup displaying structural asset descriptions directly above localized coordinate text variables.

selects a landmark, an interactive semantic overlay displays its curated historical information positioned directly above its coordinate properties. This feature transforms raw spatial data into a useful tool for tourism planners. Planners can interactively inspect the attribute data of a specific site while checking how it relates geometrically to surrounding core and border points.

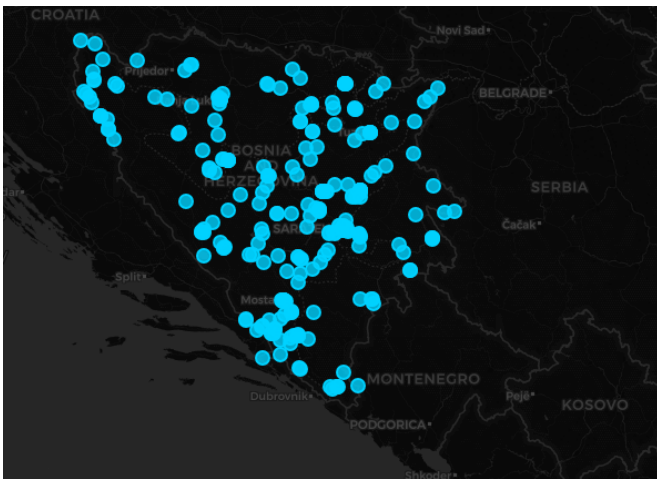


Fig. 4. Macro-corridor network map ($\epsilon = 45.0$ km) displaying structural cluster saturation across the country's main geographic paths.

Expanding the search parameters to a micro-regional tran-

sition scale ($\epsilon = 25.0$ km) merges smaller local groupings across adjacent municipalities, reducing the total cluster count to 7 and reducing isolated noise points to 6. At the macro-corridor scale ($\epsilon = 45.0$ km, $MinPts = 4$), shown in Figure 4, the entire spatial layout collapses into a single continuous tourism corridor stretching across the geographical topology of the country. Under these macro parameters, isolated spatial noise drops down to 0 outliers.

From a regional planning perspective, analyzing the points classified as spatial noise ($c = -1$) under the optimal 15.0 km threshold provides valuable developmental insights. Instead of treating these 51 points as data anomalies or extraction errors, they are classified as “hidden gems.” These points represent geographically isolated cultural or natural landmarks—such as remote necropolises, old fortresses, or isolated waterfalls—that possess high authentic heritage value but lack spatial connection to existing high-density corridors. By identifying these destinations explicitly, sustainable tourism frameworks can design targeted infrastructure policies to distribute incoming visitor traffic away from crowded city centers and into under-developed rural economies.

IV. CONCLUSION

This seminar work applied a density-based spatial clustering pipeline to map and analyze cultural tourism in Bosnia and Herzegovina. By using the Haversine metric instead of flat Euclidean distance and adding a boundary tracking check, the software calculated accurate distances near national borders and kept the spatial data clean across all 272 landmark coordinates.

The testing showed that while a radius of $\epsilon = 15.0$ km is the best setting to find tight local tourism hubs, expanding the search distance connects these hubs into larger regional travel corridors that follow natural river valleys and roads. However, a main limitation is that the algorithm currently calculates distance in a straight line over a sphere. For real-world tourism planning, future versions of this software should replace straight-line calculations with actual road driving times and terrain data to account for the mountainous geography of the country.

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